

Hyp. passing through origin: $H: \langle m, x \rangle = 0$. Geometry - partial 2 - Ref. sheet $SO(n) = \{A \in GL(n, \mathbb{R}) \mid A^T A = I, \det A = 1\}$ or $\det A = -1$

CHAPTER 5. Columns of matrix H form an orthonormal basis $\Rightarrow H^T H = I \Rightarrow$ matrix H is orthogonal ($O(n)$)

AFFINE MAP:
 $\phi: \mathbb{A}^m \rightarrow \mathbb{A}^m, \phi(x) = Ax + b$
 A = linear part (matrix)
 b = translation part (vector)
 $\exists \phi^{-1} \Leftrightarrow \det A \neq 0$

INVERSE OF THE AFFINE MAP
 $\phi(x) = Ax + b = y$
 $\Rightarrow$ take out x . $\Rightarrow \phi^{-1}(x) = A^{-1} \cdot x - A^{-1} \cdot b$

HOMOGENEOUS MATRIX of ϕ
 $H_\phi = \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix}$
 $H_\phi^{-1} = \begin{bmatrix} A^{-1} & -A^{-1}b \\ 0 & 1 \end{bmatrix}$ (rotates vectors in the yz -plane)

COMPOSITION
 $\phi_1(x) = A_1 x + b_1$
 $\phi_2(x) = A_2 x + b_2$
 $(\phi_2 \circ \phi_1)(x) = \phi_2(\phi_1(x)) \dots$
 $\Rightarrow$ another affine map

TRANSFORMATION TYPES:
 $\phi\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = A \begin{bmatrix} x \\ y \end{bmatrix} + b$
 $\downarrow$
 $AA^T = I_2$
 yes $\det A = 1$ $\Rightarrow$ direct isometry
 no (stop) $\det A = -1$ $\Rightarrow$ indirect isometry

id. $\rightarrow$ translation $\rightarrow$ rotation
 position vector $\rightarrow$ origin $\rightarrow$ direction vector $\rightarrow$ normal vector
 $B = A$
 $\downarrow$
 $A \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
 (not around origin)

AFFINE MAP: $[\phi(P)]_k = A \cdot [P]_k + b$
HOMOTHEY: $[\phi_{c,\lambda}(P)]_k = \lambda \cdot [P]_k$
 $\hookrightarrow$ rescales the space with factor λ

BASIC: $v \parallel m \Rightarrow v = \lambda m$ ($m \cdot d = 0$ (normal direction))
Hyperplane: $H: a_1 x_1 + \dots + a_m x_m = 0$
Hyperplane equation: $a^T P + a_{m+1} = 0$
Normal vector: $(a_1, \dots, a_m)$
Line passing through a point with a given direction: $l: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_p \\ y_p \end{bmatrix} + t \cdot \begin{bmatrix} v_x \\ v_y \end{bmatrix}$ (parametric form)
Symmetric matrix: $A = A^T$; **Orthonormal basis:** $A^T A = I$
Orthogonal matrix: $A^T A = I$; $O(n)$, $\det A = \pm 1$

Rotation with angle θ around O
 after a reflection in the x -axis
 $\Rightarrow$ reflection in line $y = \tan(\theta/2)x$.

EULER-RODRIGUES:
 $Rot_{u,\theta}(x) = \cos \theta x + \sin \theta (u \times x) + (1 - \cos \theta) \langle u, x \rangle u$
 rotation of angle θ and axis (Ru)
ROTATION AROUND THE ORIGIN w ANGLE θ
 $\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$

Rotations around coord. axes:
 $Rot_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$ (around Ox)
 $Rot_{y,\theta} = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$
 $Rot_{z,\theta} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

CYCLOIDS ($\mathbb{E}^2$)
 Homog. matrix of a rotation around $C(c_1, c_2)$:
 $\begin{bmatrix} \cos \theta & -\sin \theta & -c_1 \cos \theta + c_2 \sin \theta + c_1 \\ \sin \theta & \cos \theta & -c_1 \sin \theta - c_2 \cos \theta + c_2 \\ 0 & 0 & 1 \end{bmatrix}$

Characterization of affine transformations
 $\phi: \mathbb{E}^m \rightarrow \mathbb{E}^m$ is an affine transformation $\Leftrightarrow$
 $\begin{cases} \phi \text{ inj.} \\ \phi \text{ preserves lines} \\ \phi \text{ preserves oriented ratio on lines} \end{cases}$
Rotation around C (not origin):
 $Rot_\theta(x - c) + c = Rot_{\theta, c}(x)$

HOMOGENEOUS COORDINATES
 $P(p_1, \dots, p_m) \Rightarrow (p_1, \dots, p_m, 1)$
HOMOGENEOUS MATRIX
 $\phi: \mathbb{A}^m \rightarrow \mathbb{A}^m, \phi(x) = Ax + b$
 $\hat{H}_{K', K}(\phi) = \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix}$
 composition of affine maps $\Rightarrow$ multiplication of homog. matrices

$\lambda = 1 \Rightarrow$ IDENTITY MAP
 $\lambda < 1 \Rightarrow$ CONTRACTION
 $\lambda > 1 \Rightarrow$ EXPANSION

Special orthogonal matrix: $A^T A = I$; $SO(n)$; $\det A = 1$
 $\Rightarrow A = R_\theta \dots$ **Eg. of a line passing through $(0, 1)$:** $y = kx + 1$
 $[x, y, 1]^T \rightarrow$ POINT (specific location, ex: origin $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$)
 $[x, y, 0]^T \rightarrow$ VECTOR (direction / point at ∞)

ISOMETRY = map that preserves dist.
 $\phi: \mathbb{E}^m \rightarrow \mathbb{E}^m, d(\phi(P), \phi(Q)) = d(P, Q)$
 Isometries are affine transformations
 Translation maps are isometries.

ROTATION MATRIX (counterclockwise usually)
 $A \in SO(2) \Leftrightarrow A = R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
 $(\mathbb{E}^2, \mathbb{E}^3)$ Direct isometry that fixes a point $\Rightarrow$ IDENTITY or ROTATION.

DIRECT ISOMETRY:
 - identity
 - translation
 - glide-rot / -rotation

INDIRECT IS:
 - reflection
 - glide-refl.
 - roto-refl.
GLIDE REFLECTION in a line:
 = composition of an orthogonal reflection in l and a translation in the direction of l .
 $x \mapsto Ref_{\text{for}}^\perp(x) + \lambda i = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} \lambda \\ 0 \end{bmatrix}$

GLIDE ROTATION
 = composition of a rotation and a translation parallel to the rotation axis.

PARALLEL PROJ. ON HYPERPLANE
 line l , dir. vector v , point P .
 $\Rightarrow$ line l_p passing through P with dir. vector v .
 $\Rightarrow P' = l \cap l_p \Rightarrow [P']_k = \dots \begin{bmatrix} x' \\ y' \end{bmatrix}$
 $\Rightarrow Pr_{H,v}: \mathbb{A}^m \rightarrow \mathbb{A}^m$

$Pr_{H,v}(P) = \left(I_m - \frac{v \cdot a^T}{v^T \cdot a} \right) P - \frac{a_{m+1}}{v^T \cdot a} \cdot v$
 $\downarrow$ if orthonormal
 $\begin{bmatrix} \phi(x) \\ 1 \end{bmatrix} = \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} \quad \left(I_m - \frac{v \otimes a}{\langle v, a \rangle} \right)$

ORTHOGONAL PROJ. ON HYPERPLANE
 dir. vector v is orthogonal to H (normal)
 $Pr_H^\perp(P) = \left(I_m - \frac{a \otimes a}{|a|^2} \right) P - \frac{a_{m+1}}{|a|^2} \cdot a$
 $v = a$ = normal vector of H

ORTHOGONAL PROJ. ON HYPERPLANE
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 $v = a$ = normal vector of H

ELLIPSE (quadratic curve, 2D) $\rightarrow$ sum of distances to foci is constant.

$$\mathcal{E}: \{P \in \mathbb{R}^2 \mid d(P, F_1) + d(P, F_2) = 2a\}, (a) a > 0.$$

Fix frame $\Rightarrow \left[\mathcal{E}: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \right]$ canonical equation.

($\Delta < 0$)

$y = Kx \pm \sqrt{a^2 K^2 + b^2}$ (tg with slope K) $b^2 = a^2 - c^2$

$T_{(x_0, y_0)} \mathcal{E}_{a,b}: \frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$ (tg at given point) $c^2 = a^2 - b^2$

(conic section)

HYPERBOLA $\rightarrow$ abs. value of difference of distances to foci is constant. asymptotes

$$\mathcal{H}: \{P \in \mathbb{R}^2 \mid |d(P, F_1) - d(P, F_2)| = 2a\}, (a) a > 0 \quad y = \pm \frac{b}{a}x$$

Fix frame $\Rightarrow \left[\mathcal{H}: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \right]$ canonical equation. doubly ruled surface

$y = Kx \pm \sqrt{a^2 K^2 - b^2}$ (tg with slope K) $c^2 = a^2 - b^2$

$T_{(x_0, y_0)} \mathcal{H}_{a,b}: \frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$ (tg at given point)

(conic section)

PARABOLA $\rightarrow$ equidistant from focus point and direction

Fix frame $\Rightarrow \left[\mathcal{P}_p: y^2 = 2px \right]$ canonical equation

($\Delta = 0$)

$y = Kx + \frac{p}{2K}$ (tg with slope K) $x = -\frac{p}{2}$ directrix

$T_{(x_0, y_0)} \mathcal{P}_p: yy_0 = p(x + x_0)$ (tg at given point)

CIRCLE: $x^2 + y^2 = r^2$; $(x - x_0)^2 + (y - y_0)^2 = r^2$ (conic section)

HYPERQUADRIC IN $\mathbb{R}^2$ = quadratic curve = conic

$C: ax^2 + bxy + cy^2 + dx + ey + f = 0$ PARALLEL REFLECTION in H

line l , dir. vector v , point P .

more about ELLIPSE: $\Rightarrow$ line lp passing through P with d. vector v .

$b^2 = a^2 - c^2$ $\Rightarrow l \cap lp = H_{P, v} = \frac{P + P'}{2} = P_{l, v}(P) \Rightarrow P' = \dots$

$\mathcal{E}_{a,b} \cap Ox, Oy \rightarrow (\pm a, 0), (0, \pm b) \Rightarrow \text{Ref}_{H, v}: \mathbb{A}^m \rightarrow \mathbb{A}^m$

Eccentricity: $\varepsilon = \frac{c}{a} = \sqrt{1 - \frac{b^2}{a^2}} \in [0, 1)$ $P' = 2P_{l, v}(P) - P$

$H \in \mathcal{E}_{a,b} \Leftrightarrow (\pm x_H, \pm y_H) \in \mathcal{E}_{a,b}$ $\text{Ref}_{H, v}(P) = \left(I_m - 2 \frac{v \cdot v^T}{v^T \cdot v} \right) P - 2 \frac{a_{m+1}}{v^T \cdot a}$

I Parametrization: $y(x) = \pm \frac{b}{a} \sqrt{a^2 - x^2}$

Northern param: $\phi: [-a, a] \rightarrow \mathbb{R}^2$ If plane passes through origin, translation part disappears.

$\phi(x) = (x, y(x)) = (x, \frac{b}{a} \sqrt{a^2 - x^2})$ doubly ruled surface

II Parametrization: $\phi: \mathbb{R} \rightarrow \mathbb{R}^2$ $\phi(t) = (a \cos t, b \sin t)$

ELLIPTICAL MOTION: $t \mapsto \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

more about hyperbola: $b^2 = c^2 - a^2$

$\mathcal{H}_{a,b} \cap Ox, Oy \rightarrow (\pm a, 0)$

Eccentricity: $\varepsilon = \frac{c}{a} \in (0, 1)$

I Param: $y(x) = \pm \frac{b}{a} \sqrt{x^2 - a^2}$

Northern param: $\phi(x) = (x, y(x)) = (x, \frac{b}{a} \sqrt{x^2 - a^2})$

RULED SURFACE (S)

(*) $p \in S, \exists l$ st. $\begin{cases} p \in S \\ l \subseteq S \end{cases}$

$S = \bigcup_{l \in \mathcal{L}} l$

$\hookrightarrow$ generator lines of S .

ORTHOGONAL REFLECTION

$\text{Ref}_{H, v}^{\perp}(P) = \left(I_m - 2 \frac{a \otimes a}{|a|^2} \right) P - 2 \frac{a_{m+1}}{|a|^2} a$

"Point fixed by both reflections"

$\Rightarrow \pi_1 \cap \pi_2$

more about PARABOLA:

focus point: $F(\frac{p}{2}, 0)$

directrix: $d: x = -\frac{p}{2}$

$\mathcal{P}_p \cap Ox, Oy \rightarrow (0, 0)$

$H \in \mathcal{P}_p \Leftrightarrow (x_H, \pm y_H) \in \mathcal{P}_p$

(quadric) ELLIPSOID (hyperquadric surface, 3D)

$\mathcal{E}_{a,b,c}: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ (can. equation)

$T_p \mathcal{E}_{a,b,c}: \frac{xx_p}{a^2} + \frac{yy_p}{b^2} + \frac{zz_p}{c^2} = 1$ (tg plane at P)

(quadric)

ELLIPIC CONE

$\mathcal{C}_{a,b,c}: \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$ (can. eq.)

$T_p \mathcal{C}_{a,b,c}: \frac{xx_p}{a^2} + \frac{yy_p}{b^2} - \frac{zz_p}{c^2} = 0$ (tg plane at P)

(quadric) HYPERBOLOID OF ONE SHEET

$\mathcal{H}'_{a,b,c}: \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ (can. eq.)

$T_p \mathcal{H}'_{a,b,c}: \frac{xx_p}{a^2} + \frac{yy_p}{b^2} - \frac{zz_p}{c^2} = 1$ (tg plane at P)

Family 1:

$l_{\lambda, \mu}: \begin{cases} \lambda \left(\frac{x}{a} - \frac{z}{c} \right) = \mu \left(1 - \frac{y}{b} \right) \\ \mu \left(\frac{x}{a} + \frac{z}{c} \right) = \lambda \left(1 + \frac{y}{b} \right) \end{cases}$

Family 2:

$l_{\lambda, \mu}: \begin{cases} \lambda \left(\frac{x}{a} - \frac{z}{c} \right) = \mu \left(1 + \frac{y}{b} \right) \\ \mu \left(\frac{x}{a} + \frac{z}{c} \right) = \lambda \left(1 - \frac{y}{b} \right) \end{cases}$

(quadric)

HYPERBOLOID OF TWO SHEETS

$\mathcal{H}^2_{a,b,c}: \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = \pm 1$ (can. eq.)

$T_p \mathcal{H}^2_{a,b,c}: \frac{xx_p}{a^2} - \frac{yy_p}{b^2} - \frac{zz_p}{c^2} = \pm 1$ (tg plane at P)

(quadric)

ELLIPIC PARABOLOID

$\mathcal{P}^e_{a,b}: \frac{x^2}{a} + \frac{y^2}{b} - 2z = 0$

$T_p \mathcal{P}^e_{a,b}: \frac{xx_p}{a} + \frac{yy_p}{b} - 2p - 2 = 0$

(quadric)

HYPERBOLIC PARABOLOID

$\mathcal{P}^h_{a,b}: \frac{x^2}{a} - \frac{y^2}{b} - 2z = 0$

$T_p \mathcal{P}^h_{a,b}: \frac{xx_p}{a} - \frac{yy_p}{b} - 2p - 2 = 0$

Family 1:

$l_{\lambda, \mu}: \begin{cases} \lambda \left(\frac{x}{\sqrt{a}} - \frac{y}{\sqrt{b}} \right) = 2\mu z \\ \mu \left(\frac{x}{\sqrt{a}} + \frac{y}{\sqrt{b}} \right) = \lambda \end{cases}$

Family 2:

$l_{\lambda, \mu}: \begin{cases} \lambda \left(\frac{x}{\sqrt{a}} + \frac{y}{\sqrt{b}} \right) = 2\mu z \\ \mu \left(\frac{x}{\sqrt{a}} - \frac{y}{\sqrt{b}} \right) = \lambda \end{cases}$

I Param: $y(x) = \pm \sqrt{2px}$

Northern: $\phi(x) = (x, y(x)) = (x, \sqrt{2px})$

II param: $\phi(y) = (x(y), y) = \left(\frac{y^2}{2p}, y \right)$